



Statistical Profiling in the Health Insurance Industry

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1 Introduction

Private health insurance companies in various countries are allowed to vary their services and pricing options based off various information about the customer and the risks involved. The so-called “risk factors” involved when profiling include age, gender, race, sexuality, smoking habits, income, body mass index, medical history and more (Boyer-Kassem & Duchêne, 2020). Certain forms of profiling have already been banned in parts of the world¹, while others are still common. As the high price of healthcare services remains a big issue across the world², the implications of statistical profiling are an important subject of discussion. In this article, we aim to analyse this practice and its various implications through both an ethical and a policy-based framework.

2 Methodology

The question of whether companies should be allowed to vary pricing based on statistical profiling is a very broad one. Naturally, we will analyze the issue by looking at various subsets of the risk factors involved in profiling. Such subsets can be defined based on criteria such as whether the agent is

¹For example genetic profiling is currently forbidden in the US (*Genetic Information Nondiscrimination Act (GINA) of 2008*, 2008)

²For instance, nearly 2 million people declare bankruptcy in the US because of medical bills every year (Mangan, 2013)

responsible for the risk factor, whether the risk factor is causally linked to health costs, and whether the agent can be reasonably expected to know about the aforementioned causal link.

Before we proceed, we will outline a few examples of risk factors commonly used for statistical profiling that fall onto the different sides of each of our criteria. For the first criterion, age is a risk factor the agent is not responsible for, while their body mass index is one they usually are. For the second criterion, people who smoke are more likely to develop lung cancer (Varghese & Muntode Gharde, 2023), leading to high medical costs (de Nijs, de Koning, van der Aalst, & Ten Haaf, 2024), hence the individual’s smoking habits represent a risk factor that is causally responsible for the increased health costs. On the other hand, a person’s place of living could be used for profiling if correlation was observed with health costs, all without requiring the causality link to exist. As for the last criteria, while people in developed countries can, in the 21st century, be reasonably expected to know about the health risks of smoking (Ahluwalia et al., 2018), this would not have been the case a hundred years ago³.

The agents at play in our analysis are the insurance companies, and the individuals who pay for the insurance.

2.1 The ethical framework

We will analyse the ramifications of the different risk factors through two frameworks: an ethical-based and a policy-based one. For our ethical analysis, we will consider the ethical theories of consequentialism and deontology. Our analysis is primarily based on (Boyer-Kassem & Duchêne, 2020).

In particular, we will use utilitarianism (a form of consequentialism). According to (Rachels & Rachels, 2022, Chapter 7), utilitarianism is based around the “Principle of Utility”, that is, the principle that all actions should attempt to maximize the so called “utility” produced. Intuitively, our analysis through the lens of utilitarianism will focus on the consequences profiling using different classes of risk factors has on the parties involved.

Second, we will analyse the issues through the lens of deontology. Kant’s ethics (i.e. deontology) are based around the idea that humans should “always be treated as an end and never as a means only”, which he formulate into a categorical imperative (a law of morality, which all actions must abide by): “Act so that you treat humanity, whether in your own person or in

³According to (Parascandola, 2001), the turning point regarding research into the effects of smoking happened in the 1950s, thus people would have had no reasonable way to know about the causal link before then.

that of another, always as an end and never as a means only” (Rachels & Rachels, 2022, Chapter 10). A more concrete equivalent formulation of this imperative, as stated by (University of Groningen, 2025, Section 4.2.1) is the following: ”Act only according to that maxim whereby you can, at the same time, will that it should become a universal law”.

Using Kant’s theory to analyse statistical profiling turns out to be quite challenging, as making precise arguments that go beyond the fact that, in some vague sense, health insurance companies treat humans as mere data points when performing statistical profiling turns out to be no easy task. We will instead turn our attention to a specific form of deontology — *contractualism*. In (Scanlon, 1998, Page 153), Scanlon summarises the underlying principle as follows: “An act is wrong if its performance under the circumstances would be disallowed by any set of principles for the general regulation of behaviour that no one could reasonably reject as a basis for informed, unforced, general agreement”. We will thus analyse whether statistical profiling (or the lack thereof) violates any such principles.

Moreover, we will also make use of John Rawls’s theory of contractualism, which is centered around the famous *veil of ignorance*. According to (Ashford & Mulgan, 2018, Section 2), “Rawls places the parties to his agreement behind a veil of ignorance, where they do not know many key facts about their own identity. This is to ensure that the resulting principles of justice embody Rawls’s commitment to liberal neutrality”. Rawls’s contractualism is an useful form of analysis, as it puts forth the inherent anti-symmetry introduced by statistical profiling and its consequences.

2.2 The policy framework

According to (University of Groningen, 2025), “policy is a preferred course of action, with the desired consequence of solving a problem”. THIS IS VERY INCOMPLETE

3 Ethical analysis

3.1 Consequentialism

We start our analysis by looking at the questions from the perspective of utilitarianism. Our analysis is based on (Boyer-Kassem & Duchêne, 2020).

At a first glance, profiling does not seem to have an impact on the total utility of the agents at play. Indeed, if the health insurance company at hand is simply redistributing risk among the customers without making a profit

(i.e. the company is only moving money around), then whatever utility is lost by some agent by having to spend more would then be gained by everyone else having to spend a bit less.

The analysis stays the same if a percentage of the incoming revenue is taken by the company as profit or running costs. Of course, this assumes the utility of the customers is a linear function with respect to the amount of money spent. Were it the case this function was concave, then utilitarianism would side against profiling (this is easy to see in the trivial case where all the individuals have equal income). Conversely, utilitarianism would side with profiling in the case of convex utility functions. According to (Kahneman & Tversky, 1979), the utility functions observed in practice vary from individual to individual, thus we will avoid the issue by simply assuming linear functions.

In practice, statistical profiling requires more money — data collection, statistical analysis and the disputes involved introduce additional costs for the insurance companies, costs which must then in turn be distributed to the customers. Thus, if the total amount of money required for the functioning of the system is the only utility source considered for the analysis, then utilitarianism comes out against profiling.

Of course, the total amount of money spent is hardly the only matter worth taking into consideration. For one, people who have to pay more because of circumstances outside their control might feel discriminated against or stigmatized. Moreover, at the individual level, is the exchange of utility equivalent? That is, does a person who get sick but gets reimbursed by the insurance company remain utility-neutral on the matter? Does a person who is not sick, but keeps paying for insurance, gain enough utility from the sense of security received in order to counteract the loss of utility from having to spend a given amount of money?

Indeed, when looking at a company that profiles its customers, a person that has greater-than-average risk will produce additional negative utility not only when getting sick (this is always the case, be it with or without profiling), but also from potential feelings of discrimination or from feeling stigmatized. If the cause of the increased risk is one the person can influence, then the distress caused by the loss of utility could act as an additional push for the agent at hand to improve their condition, generating positive utility in the long term, although as discussed in the previous sections, many of the risks considered do not fall into this category. On the other hand, in a world without statistical profiling, people who have smaller-than-average risk will produce additional negative utility from feeling that they are paying for other people's health. Weighting the two sources of negative utility against each other without imposing further restrictions on the exact values of the utilities at hand is impossible. Since utilitarianism is, to some extent, blind to the

differences between the different risk factors, we will move our attention to analysing the issue through other ethical theories.

3.2 Deontology

Rawls uses his famous *veil of ignorance* to argue that rational agents agree on principles of justice that ensure the best outcome for those that are currently in the worst position. Intuitively, if one cannot tell whether they will end up in poor or good health, then they would rationally only favour inequalities which improve everyone's situation. Statistical profiling fails this test, as it requires those in poor situations to pay more, thus sending them a downwards spiral of degrading circumstances, since wealth and health are strongly positively correlated (Marmot, 2005). Indeed, under Rawls's contractualism, one should then side against profiling, since they might end up on the poor end of the spectrum once the veil is lifted up.

Scanlon argued that moral principles are those that cannot reasonably be rejected: "In order for a principle to be reasonably rejectable there must be some relevant standpoint from which people typically have good reason" (Scanlon, 1998, Page 218). Of course, those in poor conditions can reasonably reject the principle of allowing health insurance companies to perform statistical profiling, by bringing up arguments like the fact it puts a larger burden on those who already find themselves in terrible situations. On the other hand, healthy people wouldn't be able to reasonably make the converse argument for supporting statistical profiling, since they do not find themselves in the same terrible situations.

Of course, our analysis so far has been quite rough. In reality, the arguments' effectiveness varies greatly based on the specific risk factor being used for profiling. For one, an unhealthy person would have a hard time reasonably rejecting statistical profiling if they themselves are responsible for their poor health condition, thus making profiling permissible under Scanlon's theory. One could for example similarly argue under Rawls's theory that a person who smokes could reasonably agree even when under the veil of ignorance that they deserve to pay more for their insurance, regardless of whether they are in fact sick once the veil is lifted up.

Can we then conclude that statistical profiling is permissible, perhaps even preferable, for risk factors the individual is responsible for? As (Boyer-Kassem & Duchêne, 2020, Section 3) discusses, the issue of responsibility is a lot more complicated than it might sound. We define *responsibility* the same way as the aforementioned paper: "an agent is responsible for X if X is a result, foreseeable by the agent, of a free choice of theirs". We will first consider risk factors that fall under the intersection of the three criteria

defined in Section 2. Examples of risk factors which fall under this umbrella include one's body-mass index, smoking habits, and alcohol consumption.

The main complication when it comes to the three risk factors mentioned above comes from the fact they are usually bound to addictive behaviours. For one, those behaviours are influenced by both environmental factors (Agrawal & Lynskey, 2008), and genetic predispositions (Kerns et al., 2005). This suggests practical difficulties in assessing what percentage of one's risk factors they bear the responsibility for, especially when multiple of the aforementioned factors compound.

It now remains to look at risk factors which fail to satisfy either of the last two of our three criteria (we are still discussing things in the context of responsibility, thus we treat the first criteria as always being required). If the risk factor at hand is not causally linked to the increase in health cost, then one could reasonably argue they might be responsible for the risk factor itself, but not for the health cost incurred downstream, thus siding against profiling in Scanlon's theory. If the causal link exists, but without the individual's knowledge (e.g. in the case of smoking in the 1920s), then an affected individual can similarly argue they are not responsible for the incurred health costs, as part of our definition of responsibility was for the outcome to be *foreseeable*, leading to the same conclusion.

While there's no one definitive answer we can draw here, we conclude that under the two considered versions of contractualism, statistical profiling is to some extent permissible only when the risk factor satisfies the three outlined criteria, i.e. when the risk factor is causally linked to the health costs in a way foreseeable by the individual, while falling under the individual's responsibilities.

4 Policy analysis

Uh oh, I guess no one has written this section just yet ;-;

5 Conclusion

I am going insane :3

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Appendices

A AI usage declaration

- I, Adriel Matei, have not used any form of AI while writing this paper.
- I, Steijn van de Vliert, have used Chat GPT as a way of brainstorming about the topic we were going to pick. I also used Writefull to correct spelling and grammar. However, I have not used AI as a way of obtaining information in any way.
- TYPE YOUR DECLARATIONS HERE

B Individual contributions

- I, Adriel Matei, have written pretty much the entirety of sections 1, 2, 3.1, and 3.2.
- I, Steijn van de Vliert, contributed to finding sources and writing section 2.2
- TYPE YOUR CONTRIBUTIONS HERE